Probabilistic Combinatorics

Sheet 10

June 24, 2024

Problem 1. Formulate and proof analogs of the FKG Inequality for the cases, where

- f is monotonically decreasing and g monotonically increasing.
- f and g are both monotonically decreasing.

Proof.
By applying the FKG Inequality to the case where f is monotonically decreasing and g is monotonically increasing, then the following inequality holds

$$\sum_{x\in L} f(x)\mu(x) \sum_{x\in L} g(x)\mu(x) \geq \sum_{x\in L} f(x)g(x)\mu(x) \sum_{x\in L} \mu(x)$$

and the following inequality holds by applying the FKG Inequality to the case where f and g both monotonically decreasing

$$\sum_{x\in L} f(x)\mu(x) \sum_{x\in L} g(x)\mu(x) \leq \sum_{x\in L} f(x)g(x)\mu(x) \sum_{x\in L} \mu(x)$$

These two statements can be proved as follows.

In order to apply the FKG Inequality, suppose that L is a countable distributive lattice, $f, g : L \to \mathbb{R}$, and $\mu : L \to \mathbb{R}$ is log-supermodular, such that the following functions exist

$$\sum_{x\in L} \mu(x), \sum_{x\in L} f(x)\mu(x), \sum_{x\in L} g(x)\mu(x), \text{ and } \sum_{x\in L} f(x)g(x)\mu(x)$$

The FKG Inequality only applies to f, g that are monotonically increasing. Therefore, we want to apply the FKG Inequality to $-f$ and g when f is monotonically decreasing and g is monotonically increasing, this is because $-f$ is monotonically increasing. In this case, by the FKG Inequality, we have

$$\begin{aligned}\sum_{x\in L} \Big(-f(x)\Big)\mu(x) \sum_{x\in L} g(x)\mu(x) &\leq \sum_{x\in L} \Big(-f(x)\Big)g(x)\mu(x) \sum_{x\in L} \mu(x)\\ \Leftrightarrow \sum_{x\in L} f(x)\mu(x) \sum_{x\in L} g(x)\mu(x) &\geq \sum_{x\in L} f(x)g(x)\mu(x) \sum_{x\in L} \mu(x)\end{aligned}$$

Similarly, we want to apply the FKG Inequality to $-f$ and $-g$ when f and g are both monotonically decreasing, as $-f$ and $-g$ are monotonically increasing. In this case, by the FKG inequality, we have

$$\begin{aligned}\sum_{x\in L} \Big(-f(x)\Big)\mu(x) \sum_{x\in L} \Big(-g(x)\Big)\mu(x) &\leq \sum_{x\in L} \Big(-f(x)\Big)\Big(-g(x)\Big)\mu(x) \sum_{x\in L} \mu(x)\\ \Leftrightarrow \sum_{x\in L} f(x)\mu(x) \sum_{x\in L} g(x)\mu(x) &\leq \sum_{x\in L} f(x)g(x)\mu(x) \sum_{x\in L} \mu(x)\end{aligned}$$

, and the proof is finished. □

Problem 2. Follow from the FKG Inequality, that for real numbers $0 \leq a_1 \leq \ldots \leq a_n$ and $0 \leq b_1 \leq \cdots \leq b_n$ it holds, that

$$\langle a, b\rangle \geq \frac{\langle a, \mathbf{1}\rangle \cdot \langle \mathbf{1}, b\rangle}{n}$$

where $\mathbf{1} \in \mathbb{R}^n$ is the constant 1-vector and a and b are the vectors with ordered entries a_i resp. b_i. $\langle ., .\rangle$ is the standard dot product.

Proof.
Let $S = \{(a_1, b_1), (a_2, b_2), \ldots, (a_n, b_n)\}$ where the binary relation $\leq$ on S is defined as $(a_i, b_i) \leq (a_j, b_j)$ when $a_i \leq a_j$ and $b_i \leq b_j$. It follows from the Problem 2 in Sheet 9 that S is a distributive lattice. Next, we define the following functions for any $s_i = (a_i, b_i) \in S$

$$f(s_i) = a_i, g(s_i) = b_i, \text{ and } \mu(s_i) = 1$$

By the definition of the binary relation $\leq$ on the distributive lattice S, for $i < j$, we have $f(s_i) = a_i \leq a_j = f(s_j)$, and $g(s_i) = b_i \leq b_j = f(s_j)$. Therefore, the two functions f, g are monotonically increasing. Since $\mu(s_i) = 1$, then it is obvious that $\mu(x)\mu(y) \leq \mu(x \vee y)\mu(x \wedge y)$ for any $x, y \in S$, and thus μ is log-supermodular. Moreover, as S is a countable distributive lattice, then the following functions exist

$$\sum_{s\in S} f(s)\mu(s), \sum_{s\in S} g(s)\mu(s), \sum_{s\in S} f(s)g(s)\mu(s), \text{ and } \sum_{s\in S} \mu(s)$$

By the FKG Inequality, we have

$$\begin{aligned} \sum_{s\in S} f(s)\mu(s) \sum_{s\in S} g(s)\mu(s) &\leq \sum_{s\in S} f(s)g(s)\mu(s) \sum_{s\in S} \mu(s) \\ \Leftrightarrow \sum_{i=1}^{n} a_i \sum_{i=1}^{n} b_i &\leq n \sum_{i=1}^{n} a_i b_i \end{aligned} \tag{2.1}$$

Moreover, as $\langle a, b\rangle = \sum_{i=1}^{n} a_i b_i$, $\langle a, \mathbf{1}\rangle = \sum_{i=1}^{n} a_i$, and $\langle \mathbf{1}, b\rangle = \sum_{i=1}^{n} b_i$, the equality (2.1) is equivalent to

$$\begin{aligned} \langle a, \mathbf{1}\rangle \cdot \langle \mathbf{1}, b\rangle &\leq n\langle a, b\rangle \\ \Leftrightarrow \frac{\langle a, \mathbf{1}\rangle \cdot \langle \mathbf{1}, b\rangle}{n} &\leq \langle a, b\rangle \end{aligned}$$

, and the proof is finished. □

Problem 3. Let $p \in [0, 1]$. From the set $[n]$ there will be chosen a subset A randomly, by choosing independently with probability p for each $i \in [n]$ whether i is taken into A.

From A there will be constructed $B \subseteq A$, by removing every i with $\{i, i + 1\} \subseteq A$.

Determine the expected value of the cardinality of B and determine p such that

$$\lim_{n\to\infty} |\mathbb{E}[\#B] - \frac{n}{4}| = 0$$

Proof.
Let $(\Omega, \mathbb{P})$ be the discrete probability space where the sample space $\Omega = 2^{[n]}$ and the probability measure $\mathbb{P}(\omega) = p^{|\omega|}(1 - p)^{|n|-|\omega|}$ for any $\omega \in \Omega$. Let $X : \Omega \to \mathbb{N}$ be the random variable that assigns to any $\omega \in \Omega$ its cardinality, and $Y : \Omega \to \mathbb{N}$ be the random variable that assigns to any $\omega \in \Omega$ the number of elements $i \in [n]$ such that $\{i, i + 1\} \subseteq \omega$. That is, for any selection of A, which is $\omega \in \Omega$, then $\#B = X - Y$, and thus $\#\mathbb{B} = \mathbb{E}(X - Y)$. By linearity of expectation, we have

$$\mathbb{E}(\#B) = \mathbb{E}(X) - \mathbb{E}(Y) \tag{3.1}$$

As any subset A is constructed by independently choosing elements from $[n]$ with probability p for each $i \in [n]$, then X has binomial distribution with parameters n and p. Therefore, $\mathbb{E}(X) = np$. That is, it remains to find $\mathbb{E}(Y)$ in order to calculate $\mathbb{E}(\#B)$ by the equation (3.1).

For any $i \in [n-1]$, let $Y_i : \Omega \to \{0, 1\}$ be the indicator random variable defined by the following rule for any $\omega \in \Omega$

$$Y_i(\omega) = \begin{cases} 1 \text{ if } \{i, i+1\} \subseteq \omega \\ 0 \text{ otherwise} \end{cases}$$

Hence, $Y(\omega) = \sum_{i=1}^{n-1} Y_i(\omega)$. For $i \in [n-1]$, let $Y_i = 1_{A_i}$ be the indicator variable for the event

$$A_i = \{\omega \in \Omega \mid \{i, i+1\} \subseteq \omega\}$$

and thus $\mathbb{E}(Y_i) = \mathbb{E}(1_{A_i}) = \mathbb{P}(A_i)$. It is obvious that $\mathbb{P}(A_i) = p^2$. Therefore,

$$\begin{aligned} \mathbb{E}(Y) &= \mathbb{E}(\sum_{i=1}^{n-1} Y_i) \\ &= \sum_{i=1}^{n-1} \mathbb{E}(Y_i) \\ &= (n-1)p^2 \end{aligned}$$

Based on $\mathbb{E}(X) = np$ and $\mathbb{E}(Y) = (n-1)p^2$ that are calculated above, the equality (3.1) is equivalent to

$$\mathbb{E}(\#B) = np - (n-1)p^2$$

□